

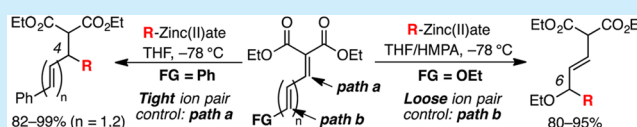
Regioselective 1,4- and 1,6-Conjugate Additions of Grignard Reagent-Derived Organozinc(II)ates to Polyconjugated Esters

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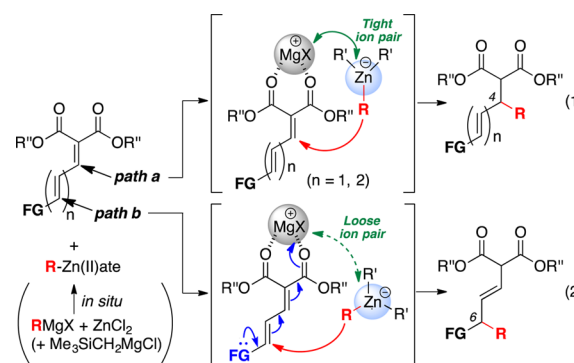
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Supporting Information

ABSTRACT: Regioselective synthetic methods were developed for 1,4- and 1,6-conjugate additions of Grignard reagent-derived organozinc(II)ates to malonate-derived polyconjugated esters. By taking advantage of the tight ion-pair control of organozinc(II)ates, it was possible to switch between 1,4- and 1,6-conjugate additions by introducing a terminal ethoxy moiety in the conjugation.



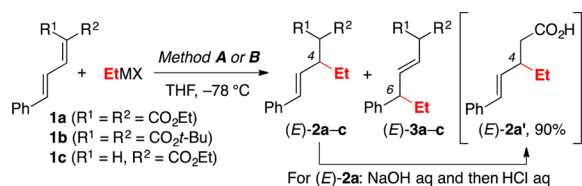
Regioselective conjugate addition to α,β -unsaturated carbon-yl compounds with organometallic reagents, which are usually prepared in situ from commercially available Grignard reagents¹ and transition-metal salts, is one of the most important carbon–carbon bond-forming reactions.² In particular, organocuprates(I) from Grignard, organozinc(II), and organoaluminum(III) reagents have been traditionally used for enantio- and regioselective 1,4-conjugate addition, and seminal breakthroughs have been achieved in the last 10 years.^{3–6} Moreover, several advanced examples of 1,6- and 1,8-conjugate additions of Grignard reagents have begun to be reported with the use of copper(I) and other transition-metal salts.⁷ Traditional organozinc(II)ates, $[\text{R}_3\text{Zn}]^-\text{[Li]}^+$ or $[\text{R}_3\text{Zn}]^-\text{[MgX]}^+$,^{8,9} have also been used for 1,4-conjugate addition of α,β -unsaturated enones,¹⁰ and recently, Dieter developed a regioselective 1,4-conjugate addition to nitrodienes.¹¹ However, the use of organozinc(II)ates for a regioselective 1,4-conjugate addition to less reactive α,β -unsaturated esters is rather limited. Unlike strongly π -coordinatable organocuprates(I) with high-lying d-orbitals, organozinc(II)ates with relatively low-lying d-orbitals would not act as metal-centered nucleophiles.^{9a,b} Alternatively, however, a tight ion pair of the hard nucleophilic $[\text{R}_3\text{Zn}]^-$ moiety and the hard Lewis acidic $[\text{Li}]^+$ or $[\text{MgX}]^+$ moiety, which would selectively coordinate to the carbonyl moiety through a pathway of carbozincation,^{9a} might be promising. Accordingly, organozinc(II)ates would recognize the carbonyl moiety better than the π -conjugated moiety, and we envisioned that organozinc(II)ates might be practical alkylating reagents for α,β -unsaturated esters, if the substrates could be designed to fully match the character of the organozinc(II)ates. In this context, we report here the transition-metal-free highly regioselective 1,4- and 1,6-conjugate additions of Grignard reagent-derived organozinc(II)ates to malonate-derived polyconjugated esters⁵ for the first time (eqs 1 and 2). In particular, chelation to the Mg(II)^+ center with a 1,3-dicarbonyl moiety would promote 1,4-conjugate addition (path a) at the activation site close to the cationic $[\text{MgX}]^+$ moiety due to the tight ion pair between the cationic $[\text{MgX}]^+$ moiety and anionic $[\text{R}_3\text{Zn}]^-$ moiety (eq 1). Moreover, if we introduce an electron-donating group to the substrates, it might weaken the tight ion pair of the organozinc(II)ates



and selectively activate the terminal conjugated carbon, so that 1,6-conjugate addition (path b) might occur selectively (eq 2).

First, we examined ethyl addition to dienyl esters **1a–c** in tetrahydrofuran (THF) at -78°C (Table 1). The reaction of diethyl malonate-derived **1a** (0.3 mmol) with ethylzinc(II)ate (1.5 equiv), which was prepared in situ from EtMgCl (1.5 equiv), $\text{Me}_3\text{SiCH}_2\text{MgCl}$ (3 equiv), and ZnCl_2 (1.5 equiv) (method A),^{8c–f} proceeded quickly, and the corresponding 4-Et adduct **2a** was obtained in 91% yield within 1 h (entry 1). On the other hand, 1.5 equiv of EtMgCl for **1a** without $\text{Me}_3\text{SiCH}_2\text{MgCl}/\text{ZnCl}_2$ (method B) provided a mixture of **2a** (56% yield) and undesired 6-Et adduct **3a** (28% yield) (entry 2). Di-*tert*-butyl malonate **1b** showed much lower reactivity than **1a**, although the ratio of **2b/3b** was still high when ethylzinc(II)ate was used (entry 3). In sharp contrast, no products were obtained from much less reactive monoester **1c** (entries 5 and 6). This result would indicate that even highly nucleophilic $[\text{Et}(\text{Me}_3\text{SiCH}_2)_2\text{Zn}]^-$ does not easily add to the nonchelatable ester **1c**, since a lack of possible chelation to the labile Lewis acid Mg(II)^+ would lead failure in keeping the nucleophilic $[\text{Et}(\text{Me}_3\text{SiCH}_2)_2\text{Zn}]^-$ close to the appropriate reaction site such as the 4-position.¹² Moreover, to avoid the synthetic limitation of malonates, hydrolysis and decarboxylation of **2a** was performed to give **2a'** in 90% yield.

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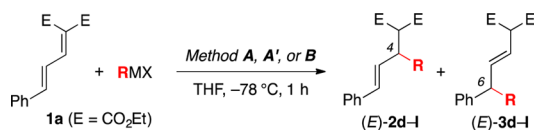
Table 1. Ethyl Addition of Dienyl Esters 1a–c^a

entry	1	method	reaction time (h)	2/3	yield (%)	
1	1a	A	1	2a/3a	91 (70) ^b	7 (26) ^b
2	1a	B	1	2a/3a	56	28
3	1b	A	24	2b/3b	71	5
4	1b	B	24	2b/3b	38	10
5	1c	A	5	2c/3c	0	0
6	1c	B	5	2c/3c	0	0

^aThe reaction was conducted in THF at -78°C with method A or B on a 0.3 mmol scale of **1**, and isolated yields are shown. Method A: EtMgCl (1.5 equiv), $\text{Me}_3\text{SiCH}_2\text{MgCl}$ (3 equiv) and ZnCl_2 (1.5 equiv). Method B: EtMgCl (1.5 equiv). ^bEtMgCl (1.5 equiv), $\text{Me}_3\text{SiCH}_2\text{MgCl}$ (0.2 equiv) and ZnCl_2 (0.1 equiv) were used.

Unfortunately, a catalytic amount of ethylzinc(II)ate was not effective, and undesired **3a** was obtained in significant amounts (entry 1, also see the Supporting Information).

We next investigated the generality of organozinc(II)ates for the 1,4-conjugate addition to **1a** (Table 2). As a result, alkyl-

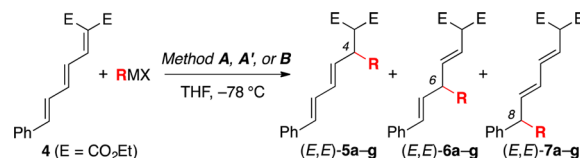
Table 2. Regioselective Alkyl Addition at the 4-Position of Dienyl Ester 1a^a

entry	RMgX	method	2/3	yield (%)	
1	MeMgCl	A'	2d/3d	85	6
2	MeMgCl	B	2d/3d	23	2
3	<i>n</i> -BuMgCl	A	2e/3e	88	10
4	<i>n</i> -BuMgCl	B	2e/3e	47	27
5	<i>n</i> -octylMgCl	A	2f/3f	90	7
6	<i>n</i> -octylMgCl	B	2f/3f	46	24
7	allylMgCl	A	2g/3g	84	9
8	allylMgCl	B	2g/3g	33	14
9	vinylMgCl	A	2h/3h	97	0
10	vinylMgCl	B	2h/3h	53	0
11 ^b	$\text{PhC}\equiv\text{CMgBr}$	A'	2i/3i	97	0
12 ^b	$\text{PhC}\equiv\text{CMgBr}$	B	2i/3i	33	0
13 ^c	PhMgBr	A'	2j/3j	99	0
14 ^c	PhMgBr	B	2j/3j	64	0
15	<i>c</i> -HexMgCl	A	2k/3k	39	60
16	<i>c</i> -HexMgCl	B	2k/3k	48	47
17 ^d	<i>t</i> -BuMgCl	A	2l/3l	25	18
18 ^d	<i>t</i> -BuMgCl	B	2l/3l	23	18

^aThe reaction was conducted in THF at -78°C with Method A, A', or B on a 0.3 mmol scale of **1a** unless otherwise noted. Method A: Grignard reagent (1.5 equiv), $\text{Me}_3\text{SiCH}_2\text{MgCl}$ (3 equiv) and ZnCl_2 (1.5 equiv). Method A': Grignard reagent (4.5 equiv) and ZnCl_2 (1.5 equiv). Method B: Grignard reagent (1.5 equiv). ^bReaction temperature was gradually increased from -78°C to -20°C over 13 h. ^cThe reaction time was 24 h. ^dThe reaction time was 20 h.

alkenyl-, and alkynylzinc(II)ates (method A or A'),¹³ which were derived from Me-, *n*-Bu-, *n*-octyl-, allyl-, and vinylMgCl and $\text{PhC}\equiv\text{CMgBr}$, could be used, and the corresponding 4-alkyl adducts **2d–i** were obtained in 84–97% yields with high

regioselectivities (entries 1–12). Moreover, phenylzinc(II)ate (method A')¹³ could also be used, and **2j** was obtained in 99% yield with a formation of a small amount (<10% yield) of biphenyl (entry 13), which is sometimes observed in the reactions with phenylzinc(II)ate.^{9j,11} Encouraged by these results, we further investigated the generality of 1,4-conjugate addition to trienyl ester **4** with 1°-alkylzinc(II)ates (Table 3).

Table 3. Regioselective Alkyl Addition at the 4-Position of Trienyl Ester 4^a

entry	RMgX	method	reaction time (h)	5/6/7	yield (%)	
1	MeMgCl	A'	3	5a/6a/7a	88	7 1
2	MeMgCl	B	3	5a/6a/7a	20	0 0
3	EtMgCl	A	1	5b/6b/7b	92	5 2
4	EtMgCl	B	1	5b/6b/7b	51	20 24
5	<i>n</i> -BuMgCl	A	1	5c/6c/7c	88	7 3
6	<i>n</i> -BuMgCl	B	1	5c/6c/7c	51	21 26
7	<i>n</i> -octylMgCl	A	17	5d/6d/7d	82	4 4
8	<i>n</i> -octylMgCl	B	17	5d/6d/7d	53	20 26
9	allylMgCl	A	1	5e/6e/7e	89	5 2
10	allylMgCl	B	1	5e/6e/7e	44	13 14
11	vinylMgCl	A	4	5f/6f/7f	87	1 1
12	vinylMgCl	B	4	5f/6f/7f	27	6 1
13	PhMgBr	A'	24	5g/6g/7g	99	0 0
14	PhMgBr	B	24	5g/6g/7g	41	0 0

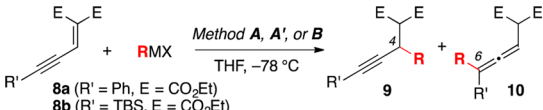
^aThe reaction was conducted in THF at -78°C with method A, A', or B on a 0.3 mmol scale of **4** unless otherwise noted.

As a result, a highly selective 1,4-conjugate addition to **4** occurred with the use of 1°-alkyl-, vinyl-, and phenylzinc(II)ate (method A or A'). In contrast, when we used Grignard reagent alone (method B), a mixture of alkyl adducts at the 4-, 6-, and 8-positions (**5**, **6**, and **7**, respectively) was obtained. However, unlike the above less hindered organozinc(II)ates, sterically demanding 2°- and 3°-alkylzinc(II)ates (*c*-Hex- and *t*-Bu, respectively) were not effective, and the desired 1,4-adducts **2k** and **2l** were obtained from **1a** with low regioselectivities (Table 2, entries 15 and 17).

To overcome this drawback in 2°- and 3°-alkyl additions to dienyl ester **1a**, we tried using less hindered enynyl ester **8a** (Table 4).¹⁴ As a result, not only 1°-alkylzinc(II)ates (entries 1 and 3) but also alkynylzinc(II)ate (entry 5), phenylzinc(II)ate (entry 7), and 2°-alkylzinc(II)ates, which were derived from *i*-Pr- and *c*-HexMgCl (entries 9, 11, and 13), could be used with method A or A'. Remarkably, 3°-alkylzinc(II)ates derived from *t*-BuMgCl could be used without serious problems (entry 15). Notably, a terminal Ph moiety (see **8a**) could be replaced by *t*-BuMe₂Si (TBS) (see **8b**), and the corresponding 4-*i*-Pr-adduct **9f** was obtained in 99% yield (entry 11).

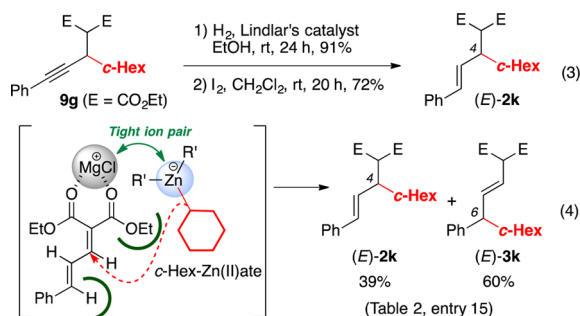
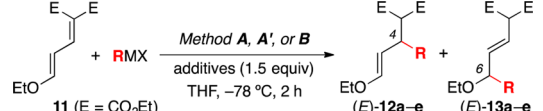
By taking advantage of the high reactivity of enynyl ester **8a**, we transformed the obtained alkynyl ester **9g** to the corresponding olefin (*E*)-**2k** (eq 3). After the hydrogenation of **9g** with Lindlar's catalyst, compound (*Z*)-**2k** was obtained in 91% yield. Subsequent treatment of (*Z*)-**2k** with iodine gave (*E*)-**2k** in 72% yield. Since (*E*)-**2k** was not successfully obtained directly from **1a** due to steric reasons (eq 4), we could demonstrate the synthetic utility of such enyne-derived products.

Finally, we investigated the reaction of dienyl ester **11**, which has a terminal EtO group¹⁵ (Table 5). As shown in eq 2, electron-donation from the EtO group would provide the most activated

Table 4. Regioselective Alkyl Addition at the 4-Position of Enynyl Esters **8a** and **8b**^a


entry	8	RMgX	method	Reaction time (h)	9/10	yield (%)
						9 10
1	8a	EtMgCl	A	1	9a/10a	92 0
2	8a	EtMgCl	B	1	9a/10a	35 0
3	8a	BnMgCl	A	2	9b/10b	94 0
4	8a	BnMgCl	B	2	9b/10b	30 3
5 ^b	8a	PhC≡CMgCl	A	14	9c/10c	98 0
6 ^b	8a	PhC≡CMgCl	B	14	9c/10c	50 0
7	8a	PhMgBr	A'	18	9d/10d	95 0
8	8a	PhMgBr	B	18	9d/10d	43 0
9	8a	<i>i</i> -PrMgCl	A	4	9e/10e	93 0
10	8a	<i>i</i> -PrMgCl	B	4	9e/10e	53 14
11	8b	<i>i</i> -PrMgCl	A	4	9f/10f	99 0
12	8b	<i>i</i> -PrMgCl	B	4	9f/10f	29 3
13	8a	<i>c</i> -HexMgCl	A	18	9g/10g	74 0
14	8a	<i>c</i> -HexMgCl	B	18	9g/10g	44 9
15	8a	<i>t</i> -BuMgCl	A	2	9h/10h	94 0
16	8a	<i>t</i> -BuMgCl	B	2	9h/10h	38 17

^aThe reaction was conducted in THF at $-78\text{ }^{\circ}\text{C}$ with method A, A', or B on a 0.3 mmol scale of **8** unless otherwise noted. ^bReaction temperature was gradually increased from -78 to $-20\text{ }^{\circ}\text{C}$ over 14 h.

Table 5. Regioselective Alkyl Addition at the 6-Position of Dienyl Ester **11** with a Terminal Ethoxy Group^a


entry	RMgX	method	additive	12/13	yield (%)
					12 13
1	EtMgCl	A	—	12a/13a	20 80
2	EtMgCl	A	HPMA	12a/13a	12 86
3	EtMgCl	A	DMI	12a/13a	14 68
4	EtMgCl	A	TMEDA	12a/13a	2 9
5	EtMgCl	B	HPMA	12a/13a	6 39
6	EtMgCl	B	—	12a/13a	13 47
7	MeMgCl	A	HPMA	12b/13b	8 80
8	MeMgCl	B	—	12b/13b	3 65
9	vinylMgCl	A	HPMA	12c/13c	2 94
10	vinylMgCl	B	—	12c/13c	4 60
11	PhMgBr	A'	HPMA	12d/13d	0 95
12	PhMgBr	B	—	12d/13d	0 28
13	<i>i</i> -PrMgCl	A	HPMA	12e/13e	10 86
14	<i>i</i> -PrMgCl	B	—	12e/13e	4 10

^aThe reaction was conducted in THF at $-78\text{ }^{\circ}\text{C}$ with method A, A', or B on a 0.3 mmol scale of **11** in the presence or absence of additive (1.5 equiv) unless otherwise noted. In method A', $\text{Me}_3\text{SiCH}_2\text{Li}$ was used in place of $\text{Me}_3\text{SiCH}_2\text{MgCl}$ in Table 5.

electrophilic carbon center at the 6-position (also see the DFT calculation in the SI). However, the possible tight ion pair of organozinc(II)ates would still promote the nearest 1,4-conjugate addition. Based on this hypothesis, we examined the use of highly coordinating additives, such as HMPA (hexamethylphosphoramide), DMI (1,3-dimethyl-2-imidazolidinone), and TMEDA (*N,N,N',N'*-tetramethylethylenediamine),¹⁶ in the presence of organozinc(II)ates to weaken the tight ion pair (Figure 1). As a

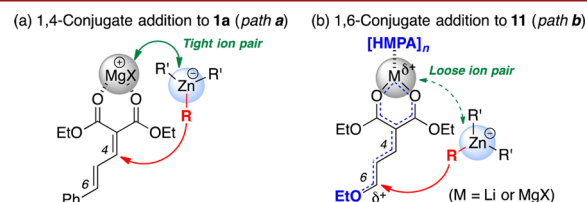
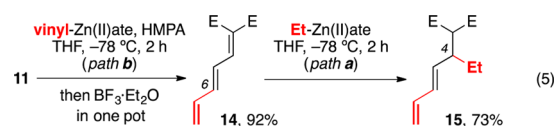


Figure 1. Control of regioselectivity with organozinc(II)ates.

result, HMPA (entry 2) was better than the other additives (entries 3 and 4).¹⁷ Desired 6-Et adduct **13a** was obtained in 86% yield with 4-Et adduct **12a** in 12% yield (entry 2), whereas **13a** was obtained in 80% yield with **12a** in 20% yield in the absence of additives (entry 1). Since EtMgCl with HMPA did not switch the regioselectivity (entries 5 and 6), the combined use of ethylzinc(II)ate and HMPA should be necessary. Other organozinc(II)ates derived from Me- and vinylMgCl could be used with method A (or A') with HMPA, and high **13** selectivities were observed (entries 7 and 9). Moreover, phenylzinc(II)ate and isopropylzinc(II)ate provided products **13d** and **13e** in respective yields of 95% and 86%.



Moreover, to demonstrate the utility of this synthetic method, we conducted a sequential path b–path a reaction (eq 5). First, compound **11** was transformed to a prolonged conjugate triene **14** with vinyl-Zn(II)ate/HPMA (path b, 1,6-addition) and the subsequent use of $\text{BF}_3 \cdot \text{Et}_2\text{O}$ in one pot. Next, compound **14** could be used for a new 1,4-conjugate addition (path a), and 4-Et-adduct **15** was selectively obtained as a major product.

In summary, we have developed highly regioselective 1,4- and 1,6-conjugate additions of Grignard reagent-derived organozinc(II)ates to malonate-derived polyconjugated esters. The tight ion-pair control of organozinc(II)ates could promote 1,4-conjugate addition to dienyl, trienyl, and enynyl esters. Moreover, a terminal activating EtO moiety in the substrate with the use of HMPA could lead to 1,6-conjugate addition successfully. Overall, the scope of substrates can prove the efficient methodology for the regioselective formation of C–alkyl, C–aryl, C–alkenyl, and C–alkynyl bonds.

■ ASSOCIATED CONTENT

§ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.orglett.6b01774.

Experimental procedures and characterization data (PDF)

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Notes

The authors declare no competing financial interest.

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- (14) The alkyne moiety was found to be highly effective for relieving the steric constraints in alkyl addition to α -imino esters. See ref 8e.
- (15) For synthetic reasons, the EtO group was selected. We examined ethyl addition to substrates with other groups, such as MeO, *i*-PrO, $\text{Me}_3\text{SiCH}_2\text{O}$, and *N*-pyrrolidinyl. However, due to the stability of these starting materials and/or the corresponding products, a complex mixture was obtained. See the SI.
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